

Capstone Weekly Status Report 14

Date and Time: December 7, 2025

Student Name: Satwik Alla

Capstone Project: SBOM Manager - Software Bill of Materials Management System

Capstone Sponsor: Software Engineering Department **Ethical Software Lab** – In-house project

Capstone Advisor: Prof. Mohammad Samarah & Prof. Melissa Sahl

Status Details:

Work accomplished this Week

Implementation & Enhancement Phase

Incorporated all Beta checkpoint feedback and implemented recommended improvements. Added progress indicators for large file processing operations and advanced filtering capabilities including platform, license type, and component count filters. Developed bulk operation functionality for simultaneous management of multiple applications. Strengthened error messaging and recovery mechanisms throughout the system. Completed vulnerability integration with automated CVE alert notifications and finalized Docker containerization with pre-installed Syft. Implemented comprehensive end-to-end testing suite and added real-time WebSocket updates for live data synchronization.

Validation & Documentation Phase

Conducted comprehensive security and performance testing across all system components. Verified cross-browser compatibility and performed user acceptance testing with external testers. Completed final code cleanup and documentation. Prepared complete Final Capstone Report (40-50 pages) including executive summary, final presentation slides, and demo video (10-15 minutes). Compiled and organized all 13+ final checkpoint deliverables for submission.

Work Planned for Next Week

Final Week: Completion & Presentation

Addressed Beta feedback and added progress indicators, advanced filtering, and bulk operations. Enhanced error messaging and completed vulnerability integration with CVE alerts. Finalized Docker containerization and implemented real-time updates. Conducted comprehensive testing and final code cleanup. Prepared Final Capstone Report (40-50 pages), presentation slides, demo video (10-15 minutes), and all 13+ deliverables for Monday submission and Wednesday presentation.

Problems, obstacles, needs, or questions that I need help with from client, advisor, or instructor

None

Overall Status

- Project Status Color: **Green**
- Project Status Color from Previous Week: **Green**